Grant application form 2022

NWO Talent Programme - Rubicon

1. General information

1a. Title of research proposal

Communication in the brain: flexible signalling with fixed lines

1b. Scientific summary of research proposal

Distant brain areas are connected through neurons that make long-range connections. Nearly all brain functions involve shifting interactions between areas and thus flexible control of the activity of neurons projecting to a given area. Although technological advances in multi-area recordings have sparked progress, it is particularly unclear (1) how cortical interarea communication is structurally implemented and depends on direct projections and (2) how flexible control is achieved while the neurons projecting to a given area remain relatively fixed.

We propose to relate functional interarea connectivity to structural connectivity by combining multi-area calcium imaging in the mouse visual cortex with viral labelling of projection neurons. We will use recently developed frameworks of population-based measures of interarea interactions to investigate how functional connectivity corresponds to direct connections (versus indirect connections or shared modulations). Realizing these experiments during a flexible visual processing task in mice, we further aim to study selective propagation of task-relevant information.

Our results can lead to new models of multiarea activity with empirically informed structural connectivity. This helps to interpret the function of different interarea connectivity motifs in the healthy brain, but also simulate and predict how aberrant connectivity impacts flexible communication in the context of cognitive disorders.

1c. Scientific summary of research proposal

Five keywords
1. Long-range projections
2. Inter-area communication
3. Feedforward and feedback signalling
4. Communication subspace
5. Cognitive flexibility

1d. Domain to which you submit your proposal (choose one)

Please indicate the applicable domain (☒):	☐ Domain Applied and Engineering Sciences (AES) ☒ Domain Science (S) ☐ Domain Social Sciences and Humanities (SSH) ☐ Health research and Medical sciences (ZonMw)

1e. Main field of research (only when applying to AES, S or SSH) ⓘ

	Code:	Field of research:
If applicable, code + other fields of research, in order of relevance:	40.20.00	1. Biologische en medische psychologie

1f. Public summary of research proposal

NL

Projecttitel
Dr. M.N. Oude Lohuis, Portugal, Champalimaud Foundation
Verschillende hersengebieden zijn verbonden door middel van lange-afstands verbindingen tussen neuronen. Deze verbindingen veranderen niet snel en toch vereist onze cognitive snel wisselende interacties tussen hersengebieden en flexibele communicatie. Dit project probeert uit te vogelen hoe het brein dynamisch controleert 'wie met wie' praat.
Wordcount: 44

ENG

Project title
Dr. M.N. Oude Lohuis, Portugal, Champalimaud Foundation
Nearly all brain functions depend on rapidly changing communication between different brain areas, even though connections between them are relatively fixed. The researchers will measure neural activity across areas and label the reciprocal connections to understand how areas communicate and decipher how the brain dynamically controls 'who talks to who'.
Wordcount: 50

2. Research proposal

2a. Proposed research

Cognition requires different brain areas to cooperate and exchange information. Crucial for cognitive flexibility is the ability for neurons in different brain areas to reorganize communication during different tasks. However, how this task-dependent communication between different areas occurs is only beginning to be understood. This problem has mainly been pursued in the mammalian visual system guided by extensive knowledge of its structural and functional organization. Theories of selective visual processing across areas rely on temporal coordination of spiking activity[1], coordination through oscillatory dynamics[2], and, more recently, the structure of activity at the population level[3–7]. Specifically, these latter studies promise to inform us - relative to earlier work - on how population activity across areas is coordinated and relates to ongoing representations and computations.

Within-area population activity is often found to be low-dimensional (i.e., activity across neurons is correlated and a small number of latent variables explains the covariance of many neurons)[8,9]. Communication between areas is similarly low-dimensional, meaning that activity fluctuations along only a subset of dimensions are driving a downstream area, a *communication subspace*[3]. This can be understood as a downstream area being particularly responsive to one or two activity patterns of many neurons. Signals that match the communication channel are effectively communicated (like a lock-and-key mechanism), while those that do not remain confined to the source area. These principles might underlie dynamic communication. Indeed, the communication subspace between visual areas appears to change as a function of visual inputs, suggestive of dynamic interactions between areas[4,7].

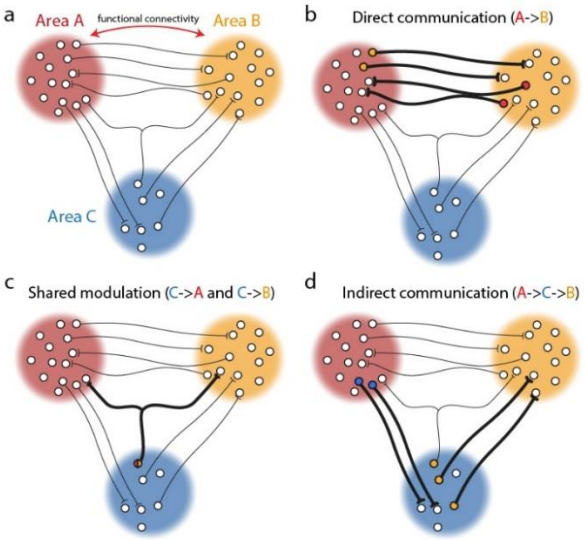


Figure 1. Anatomical routes for interarea communication. *a) Schematic of three areas connected through projection neurons. Correlated activity between two areas (here A and B) can have various anatomical origins. Communication can be the result of direct projections between area A and B (panel b), the result of shared input from area C (panel c) or occur indirectly through area C (panel d). Recording area C and knowing the identity of the projection neurons would allow to better approximate direct communication.*

Although these studies offer unprecedented insight into the nature of distributed activity, we do not know whether this indeed reflects direct communication between areas (Figure 1). Measures of functional connectivity are based on correlated activity and could reflect shared modulation by a third source. Indeed, movement and behavioural state-related signals across cortical areas lead to high values of functional connectivity, but likely reflect widespread modulation rather than direct interaction between pairs of areas[4]. Functional connectivity might further reflect *indirect* communication. For example, propagation across cortical areas might result, to a large extent, from pathways through thalamic nuclei rather than direct projections[10].

Our overall goal is to develop a novel and instructive angle on interarea communication by combining functional connectivity with knowledge of the structural connectivity (i.e., the actual long-range connections that mediate direct communication). **Our first aim** is to investigate how interarea functional connectivity relates to structural connectivity. By identifying the projection neurons, we aim to exclude shared modulations and indirect communication and better approximate what is being sent directly from area to area. **Our second aim** is to investigate task-dependent communication despite fixed anatomical projections. Although afferent synapses in the target area might rewire, the neurons projecting to a target area are stable[11]. Knowing their identity allows to investigate how their activity can be controlled to change the downstream response and implement flexible signaling. We hypothesise that activity in neurons with direct projections is specifically reorganised during flexible communication (see also analysis section).

Experimental approach

We investigate these aims in a single set of experiments that monitor the propagation and communication of visual information across the mouse cortex whilst knowing the inter-areal connectivity. The experiments involve: (a) behavioral paradigm, (b) imaging multi-area activity, and (c) labelling projections.

Behavioral paradigm - Mice will be trained in a corridor navigation task in virtual reality, which is routinely performed in the host lab. In this corridor, a zone with visual texture A is associated with reward availability, whereas texture B is not, i.e., go/nogo discrimination. Two additional stimuli (C and D) will be presented that are irrelevant to both navigation and reward availability (task design: non-informative by virtue of their location in the corridor and/or blockwise presentation structure). We hypothesize that all visual stimuli drive primary visual cortex (V1), but only the relevant stimulus pair involves extended processing and downstream communication with higher visual and associative areas to drive decision-making and motor preparation.

Imaging multi-area activity – I will perform calcium imaging of simultaneous multi-area activity in layers 2/3 of dorsal cortex of genetically modified mice expressing GCaMP-sensor. Imaging will be performed using a 2-photon microscope with a large field of view recently acquired and assembled at the Champalimaud Foundation (CF) (2P-RAM mesoscope[2]). Thanks to a large field of view (5mm x 5mm), the mesoscope allows cellular level imaging (resolution lateral <1 µm, axial <5 µm) of many dorsal cortical areas simultaneously.

We will focus on V1 and adjacent higher visual area Posteromedial (PM). PM has strong reciprocal connections with V1[12] and is, together with the retrosplenial system, involved in navigation using visual landmarks[13–15]. We hypothesize strong selective propagation between V1 and PM, given task-related modulation in medial parietal cortex [14,16,17], but this focus is reconsidered depending on observed task-related modulation of activity. Additionally, our areas of interest include retrosplenial cortex, and cingulate cortex to compare vision-related signals to motor and error-related signals[18].

Labelling projections - A viral recombination strategy will be used that labels the projections neurons between a pair of areas (Figure 2). We opt for this reciprocal labelling to investigate looped interactions (see later), compare feedforward and feedback communication, and reduce the number of animals. To label V1 neurons projecting to PM, I will deliver a retrogradely-infecting AAV expressing Cre-recombinase in PM (retroAAV-pgk-Cre) along with an AAV

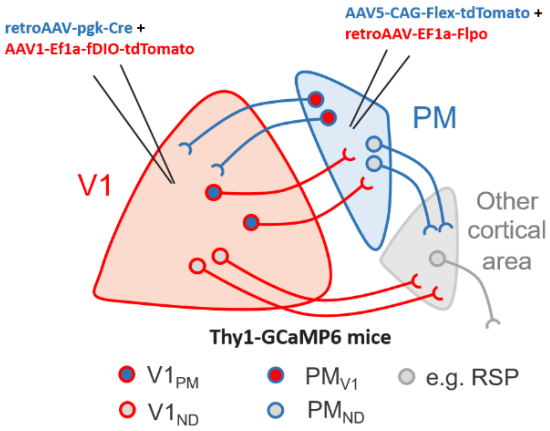


Figure 2. Viral approach to label reciprocal projections between V1 and PM. *PM neurons projecting to V1 are labelled by infecting PM with AAV5-CAG-Flex-tdTomato and V1 with retroAAV-pgk-cre (blue). V1 neurons projecting to PM are labelled by infecting V1 with AAV1-Ef1a-fDIO-tdTomato and PM with retroAAV-Ef1a-FlpO. Through conditional expression, the two sets of projection neurons will both be labelled with tdTomato. Neuronal cell bodies are schematically labelled by where they project. Thy1-GcaMP6s mice will allow monitoring calcium activity of both labelled and non-labelled excitatory cells.*

expressing cre-dependent tdTomato in V1 (AAV5-CAG-Flex-tdTomato). To label PM neurons projecting to V1, I will inject a retrograde AAV expressing Flp in V1 (retroAAV-EF1a-Flpo) and an AAV expressing Flp-dependent tdTomato (AAV1-Ef1a-fDIO-tdTomato) in PM. Such a combinatorial viral strategy using Cre and Flp recombinase technologies[19] can be successfully combined with multi-area, large scale optical recordings of neural activity[20,21].

Analysis:

For the analyses we will closely collaborate – through regular shared lab meetings and progress meetings – with Christian Machens' group at CF, which pioneered the study of interarea communication at the population level[3,5,22,23].

Aim 1: How does interarea functional connectivity relate to structural connectivity?

Functional connectivity between areas is based on correlated activity and could reflect two scenarios. It could mostly correspond to direct cortico-cortical interactions, or alternatively reflect predominantly indirect communication and shared modulations (Fig. 1). Primarily, we aim to relate the communication subspace between source and target area to the neurons projecting to the target area (labelled), or projecting elsewhere (non-labelled). We predict that non-labelled neurons will also participate in the dominate modes of correlated activity, because shared modulations will be captured as functional connectivity[4,5]. With our multi-area recordings we can distinguish global fluctuations and **predict that the subset of population activity patterns in the source area that uniquely predicts target area activity has a closer correspondence to the labelled projection neurons**. Second, if the projecting neurons capture the direct and unique interaction between the areas, does this connectivity correspond to specific task variables? In other words, does population activity that drives downstream areas relate to the stimulus, decision, or movements[4] and depend on the projection types involved? We **hypothesize that the projection subspace between V1 and PM reflects visual processing in line with recurrent processing in the visual system, rather than shared modulations (e.g. behavioral state)**. Third, our approach labels both feedforward and feedback connections (V1->PM and PM->V1). Whereas feedforward projections are known to be more driving and involved in complex feature extraction, feedback projections are thought to be more modulatory[24] and support prediction and expectation, among other cognitive functions[25–27]. Projections might thus be asymmetric. Previous work from the Machens lab showed that in anesthetized primates feedforward and feedback communication have distinguishable population activity patterns that are communicated between V1 and V2[5], but the anatomical implementation and the relevance *in vivo* is unclear. One possibility is that segregated populations of neurons form separate communication channels. However, in the host lab, it was shown that cortico-cortical connections selectively engage with neurons projecting back to them[12]. This means that both feedforward and feedback projections between V1 and PM make stronger connections with neurons that project back to the source of the input – 'looped' neurons. Because looped neurons allow for quick feedback signals, **we hypothesize that activity in the source area explains activity of labeled neurons (projecting back to the source area) better than non-labeled neurons.**

Aim 2: How is flexible interarea communication achieved given fixed connectivity?

Our behavioral task design includes task-relevant stimuli (A and B for Go/Nogo discrimination), and distractor stimuli (C and D) that are simultaneously presented but irrelevant. This allows to compare visual processing of relevant vs irrelevant stimulus trials under equal motor programs and reward expectations. We aim to investigate whether flexible gating of areal communication can be achieved by controlling within-area activity to match or not match the communication subspace[3,6,28]. We **predict that population activity in the source area (V1) is amplified along those dimensions that impact the relevant downstream target area(s) (PM) during task-relevant stimuli, but avoids those during irrelevant epochs.** In other words, activity projected along (a fixed) communication subspace would be

amplified during relevant versus irrelevant stimuli[29], for example by aligning the source-area activity to the communication subspace[30] or through uniform synaptic gain changes at the target area. An alternative hypothesis of flexible gating is that activity changes in the target area modify the impact of source activity patterns, which leads to a change in the measured communication subspace. Our simultaneous recording in combination with our ability to exclude indirect communication allows us to discriminate between these two alternatives.

In sum, this project intersects recent advances in mesoscoping imaging, viral labelling techniques, and analytical methods to relate structure and function and answer: how is neural activity routed between defined neuronal populations among a distributed network of areas. Ultimately this project aims to further our understanding of a central problem that the brain faces: how can I communicate the right thing at the right time with my neighboring area if I only have a limited set of fixed connections.

Planning, feasibility, and dissemination of the work:

Activity	Period
Pilots: behavioural task, viral labelling and multi-area imaging	**Jun 2023 – Nov 2023**
Perform experiments	**Dec 2023 – Dec 2024**
Data analysis & write paper (Aim 1)	**Sep 2024 – Jan 2025**
Dissemination through scientific meetings and conferences	**Jun 2024 – May 2025**
Data analysis & write paper (Aim 2)	**Jan 2025 – May 2025**

Table 1. Planning of research.

The applicant is skilled in working with mouse perceptual behavioral tasks, multi-area neural recordings (electrophysiology) and viral approaches in visual cortex. The particular project demands – calcium imaging and multi-area analyses – are expertises in the host institute. Moreover, in the host lab visually-guided virtual reality tasks, viral labeling and calcium imaging are all routinely performed and the mesoscope is functional.

Knowledge exchange with the scientific community will be through publications in academic journals and presentations at scientific conferences which will occur after the initial stage of the proposed research (2024-2025). We estimate two scientific publications, to be submitted around late 2024/early 2025. Communication with parties outside academia will occur more infrequently, but several popular-scientific magazines, radio programmes, newspapers and meetings are available in both Portugal and the Netherlands.

2b. Literature references ⓘ

1. Ruff, D. A. & Cohen, M. R. Attention Increases Spike Count Correlations between Visual Cortical Areas. *J. Neurosci.* **36**, 7523–7534 (2016).
2. Fries, P. A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends Cogn. Sci.* **9**, 474–480 (2005).
3. Semedo, J. D., Zandvakili, A., Machens, C. K., Yu, B. M. & Kohn, A. Cortical Areas Interact through a Communication Subspace. *Neuron* **102**, 249–259 (2019).
4. Ebrahimi, S. *et al.* Emergent reliability in sensory cortical coding and inter-area communication. *Nature* **605**, 731–721 (2022).
5. Semedo, J. D. *et al.* Feedforward and feedback interactions between visual cortical areas use different population activity patterns. *Nat. Commun.* **13**, 1099 (2022).
6. Kohn, A. *et al.* Principles of Corticocortical Communication: Proposed Schemes and Design Considerations. *Trends Neurosci.* **43**, 725–737 (2020).
7. Javadzadeh, M. & Hofer, S. B. Dynamic causal communication channels between neocortical areas. *Neuron* **110**, 2470-2483.e7 (2022).
8. Elsayed, G. F. & Cunningham, J. P. Structure in neural population recordings: an expected byproduct of simpler phenomena? *Nat. Neurosci.* **20**, 1310–1318 (2017).
9. Churchland, M. M. *et al.* Neural population dynamics during reaching. *Nature* **487**, 51–56 (2012).
10. Sherman, S. M. Thalamus plays a central role in ongoing cortical functioning. *Nat. Neurosci.* **16**, 533–541 (2016).
11. Kim, E. J. *et al.* Extraction of Distinct Neuronal Cell Types from within a Genetically Continuous Population. *Neuron* **107**, 274-282.e6 (2020).
12. Young, H., Belbut, B., Baeta, M. & Petreanu, L. Laminar-specific cortico-cortical loops in mouse visual cortex. *eLife* **10**, e59551 (2021).
13. Wang, Q., Sporns, O. & Burkhalter, A. Network analysis of corticocortical connections reveals ventral and dorsal processing streams in mouse visual cortex. *J. Neurosci.* **32**, 4386–4399 (2012).

14. Funamizu, A., Kuhn, B. & Doya, K. Neural substrate of dynamic Bayesian inference in the cerebral cortex. *Nat. Neurosci.* **19**, 1682–1689 (2016).
15. Fischer, L. F., Mojica Soto-Albors, R., Buck, F. & Harnett, M. T. Representation of visual landmarks in retrosplenial cortex. *eLife* **9**, e51458 (2020).
16. Oude Lohuis, M. N., Marchesi, P., Pennartz, C. M. A. & Olcese, U. Functional (ir)Relevance of Posterior Parietal Cortex during Audiovisual Change Detection. *J. Neurosci.* **42**, 5229–5245 (2022).
17. Pho, G. N., Goard, M. J., Woodson, J., Crawford, B. & Sur, M. Task-dependent representations of stimulus and choice in mouse parietal cortex. *Nat. Commun.* **9**, 1–16 (2018).
18. Norman, K. J. *et al.* Post-error recruitment of frontal sensory cortical projections promotes attention in mice. *Neuron* **109**, 1202-1213.e5 (2021).
19. Madisen, L. *et al.* Transgenic Mice for Intersectional Targeting of Neural Sensors and Effectors with High Specificity and Performance. *Neuron* **85**, 942–958 (2015).
20. Chen, J. L., Voigt, F. F., Javadzadeh, M., Krueppel, R. & Helmchen, F. Long-range population dynamics of anatomically defined neocortical networks. *eLife* **5**, e14679 (2016).
21. Condylis, C. *et al.* Context-Dependent Sensory Processing across Primary and Secondary Somatosensory Cortex. *Neuron* **106**, 515-525.e5 (2020).
22. Semedo, J. D., Gokcen, E., Machens, C. K., Kohn, A. & Yu, B. M. Statistical methods for dissecting interactions between brain areas. *Curr. Opin. Neurobiol.* **65**, 59–69 (2020).
23. Gokcen, E. *et al.* Disentangling the flow of signals between populations of neurons. *Nat. Comput. Sci.* **2**, 512–525 (2022).
24. D'Souza, R. D., Meier, A. M., Bista, P., Wang, Q. & Burkhalter, A. Recruitment of inhibition and excitation across mouse visual cortex depends on the hierarchy of interconnecting areas. *eLife* **5**, e19332 (2016).
25. Gilbert, C. D. & Li, W. Top-down influences on visual processing. *Nat. Rev. Neurosci.* **14**, 350–363 (2013).
26. Olcese, U., Oude Lohuis, M. N. & Pennartz, C. M. A. Sensory Processing Across Conscious and Nonconscious Brain States: From Single Neurons to Distributed Networks for Inferential Representation. *Front. Syst. Neurosci.* **12 (49)**, (2018).
27. Pennartz, C. M. A., Dora, S., Muckli, L. & Lorteije, J. A. M. Towards a Unified View on Pathways and Functions of Neural Recurrent Processing. *Trends Neurosci.* **42**, 589–603 (2019).
28. Kaufman, M. T., Churchland, M. M., Ryu, S. I. & Shenoy, K. V. Cortical activity in the null space: permitting preparation without movement. *Nat. Neurosci.* **17**, 440–448 (2014).
29. Srinath, R., Ruff, D. A. & Cohen, M. R. Attention improves information flow between neuronal populations without changing the communication subspace. *Curr. Biol.* **31**, 5299–5313 (2021).
30. Barbosa, J., Proville, R., Rodgers, C. C., Ostojic, S. & Boubenec, Y. Flexible selection of task-relevant features through across-area population gating. *BioRxiv* 2022.07.21.500962 (2022) doi:10.1101/2022.07.21.500962.

2c. Data management section

With this section, NWO mainly wants to raise awareness about the importance of responsible data management. The section is therefore not included in a committee's decision about whether or not a proposal should be awarded funding. NWO does, however, submit this section to the committee for advice.

1. Will this project involve re-using existing research data?
   - ☐ Yes: Are there any constraints on its re-use?
   - ☒ No: Have you considered re-using existing data but discarded the possibility? Why?

   If no, please briefly explain why; if yes, state any constraints on re-use of existing data if there are any.

The proposed experiments offer data that does not exist and are necessary to answer the research questions. There is one dataset that uses the technique of labeling projection neurons while attempting to image in two areas (Chen *et al.* 2016; Ref 20 in section 2a). However, in this dataset neural activity in the two areas was not acquired truly simultaneously, was restricted to two areas and thus does not permit identification of shared activity across areas, and the study did not analyze population activity nor investigate task-dependent communication.

2. Will data be collected or generated that are suitable for reuse?
   - ☒ Yes: Please answer questions 3 and 4.
   - ☐ No: Please explain why the research will not result in reusable data or in data that cannot be stored or data that for other reasons are not relevant for reuse.

3. After the project has been completed, how will the data be stored for the long-term and made available for the use by third parties? Are there possible restrictions to data sharing or embargo reasons? Please state these here.

   This proposal will generate behavioral and neural data. Behavioral data include running, licking, stimulus information and task performance, as well as videographic data of the mouse facial area. Neural data will consist of the calcium activity of 1000s of neurons across multiple cortical areas, combined with whether neurons are projecting to a given area. Research data management will follow the guidelines of NWO. Data will be deposited in a shared storage at CCU so that other members of the community can access it, in addition to being stored in a physical hard-drive for long term storage and sharing. Data will be organized using standardized neuroscience data format, NeuroDataWithout Borders (NWB) to ensure easy interoperability and foster collaboration. Any custom-programs to label and analyse data will be openly shared. When we publish, all data, code and algorithms will be uploaded to an online data repository and labelled appropriately so that it can be accessed by the entire scientific community. In addition to the FAIR data practices, we will share the data and results of the proposal with the scientific community at internal and external conferences, workshops and symposiums, in addition to open access publications and preprints.

4. Will any costs (financial and time) related to data management and sharing/preservation be incurred?

   ☐ Yes: Then please be sure to specify the associated expenses in the budget table of this proposal.

   ☒ No: All the necessary resources (financial and time) to store and prepare data for sharing/preservation are or will be available at no extra cost.

2d. Ethical aspects

Possible relevant aspects are:

- research on animals
- informed consent
- privacy and data protection
- research on human Embryonic Stem Cells
- research involving developing countries
- biosecurity/dual use.

Measures

Please provide a summary of the measures taken and measures that will be taken with regard to ethical aspects of your study.

The ethical implications of our project are those arising from research on rodents, as well as the use of genetically modified rodent models. All experimental procedures will be carried at the Champalimaud Center for the Unknown (CCU) Vivarium, which is licensed for animal experimentation by the Portuguese Directorate-General for Animal Welfare (DGAV-Direção Geral de Alimentação e Veterinária. The CCU Vivarium complies with European Guidelines (Directive 2010/63/EU of September 22, 2010), National Laws (Decree-Law 113/2013, of August 7) and Federation of European Laboratory Animal Science Associations (FELASA) guidelines and recommendations concerning laboratory animal welfare, scientific use and proper education/training of all personnel performing animal work. All experiments performed at the CCU Vivarium require prior evaluation and recommendation by the Champalimaud Animal Welfare Body and the DGAV before starting. The facility is run by a specialized team (integrating a veterinary service) that assures constant monitoring of animals during animal housing and experiments, ensuring that all handling and interventions are performed according to suitable and humane techniques following the European and institutional guidelines. The researcher Dr. Oude Lohuis has a FELASA/DGAV Function A license for carrying out procedures on animals.
We will use genetically modified GP4.3 mice (Thy1-GCaMP6s; JAX 024275) as the constitutive expression of the calcium sensor GCaMP in excitatory neurons allows to monitor neural activity across cortical areas and allows combination with recombinant-dependent viral vectors. All experiments using animals described in the proposal will be performed following Russel and Burch's 3R's – Replacement, Reduction, Refinement.
Replacement: We will use rodents, as there is no methodological alternative that will allow us to answer the questions addressed in this project. Although some invertebrate species can be used as alternatives for genetic manipulations,

their neural circuit is vastly different, and therefore cannot be used to investigate the questions regarding cortical circuits proposed here. Rodent brains offer a similar circuit organization to that of humans, and can be widely manipulated to investigate the relation between neural circuits and behaviour using molecular tools.
Reduction: In order to significantly reduce the number of mice required to obtain data in the proposed study we will (1) combine the behavioural task, viral labelling, and calcium imaging in the same mice and perform (2) mesoscopic calcium imaging techniques that allow recording the activity of many simultaneous neurons (100s-1000s). The latter not only provide us with important data on interarea communication, but also will reduce the number of animals used.
Refinement: Surgical implantation of a cranial window and viral injection are necessary to chronically monitor structurally labeled neuronal activity. Animals will be handled and constantly monitored during and after the surgery, as well as throughout the behavioral experiments such that stress and discomfort are minimized. Mice will receive analgesia throughout the surgery (Isoflurane, Lidocaine) and beyond (Buprenorphine, working action over 24h). Previous work using similar procedures has demonstrated that mice recover quickly from surgery and that implants do not affect their natural behavior. Animals will be constantly monitored during the post-surgery period until they recover. Animal welfare will be checked daily and euthanasia will be done when applicable and under deep anesthesia in a humane way.

All genetically modified mice experiments will be performed at the CCU Vivarium in compliance with the European guidelines (2009/41/EC) that was transposed to the Portuguese Law in the Decree-law 55/2015 of April 15. In particular, regarding the maintenance, storage, use, transportation, disposal and the use of physical and/or chemical/biological barriers to limit their contact with the general population and the environment. The Champalimaud Foundation follows the mandatory good laboratory practices determined by the legislation and is authorised by the competent national agency Agência Portuguesa do Ambiente (APA, I.P.) to work with genetically modified organisms and microorganisms (GMO and GMM) of risk level 1 and 2. All biological and chemical waste generated during experimental procedures will be collected and disposed by a certified company for the management of biological, chemical and hospital waste - Stericycle (Ambimed), complying with European and National Laws including the Decree-Law no. 178/2006, of 5 September which transposes Directive 2008/98 / EC of the European Parliament and of the Council of 19 November on waste, and amendments to various legal regimes in the field of waste management to mitigate any possible risk of harming the environment caused by the research. CF follows the ARRIGE (Association for Responsible Research and Innovation in Genome Editing) standards, and is part of the ARRIGE community. For environmental related issues, CF liaises with the Portuguese Environmental Agency, APA (Agência Portuguesa do Ambiente).

Ethical approval: We already submitted the full documentation for ethics revision and approval. The proposed research project will not start without the final approval by the national authorities, which will be kept on file and provided to NWO upon request.

Approval

Indicate for each approval required whether the approval has been received, applied for or not yet applied for:

	Not applicable	Not yet applied*	Applied*	Received*
Approval from a recognised medical ethics review committee	☒	☐	☐	☐
Approval from an animal experiments committee	☐	☐	☒	☐
Permission for research with the population screening Act	☒	☐	☐	☐

**You must send a copy of the applicable documents to NWO/ZonMw before your project can start*

Code of conduct

I endorse and when applicable will act accordingly:

The code of conduct for use of laboratory animals:	☒ Yes ☐ Not applicable
The code of conduct for biosecurity/possibility for dual use of the expected results:	☐ Yes ☒ Not applicable

2e. Grant budget

Costs	Amount (€)
Salary including social fees and taxes	€ 118.020
Research costs	€ 5.520
Travel costs	€ 2.000
Total	€ **125.540**

2f. Planned duration of visit (in months and fte)

Months	FTE
24	1 FTE

2g. Specify other grants

Type of grant	Amount	Source
EMBO Postdoctoral fellowship	52.800 EUR/year, 24 Months, 1 FTE	EMBO

3. Host institute

3a. Employment

Employed by a Dutch institute and seconded to a foreign Host institute	☐ Yes ☒ No *If 'Yes' please provide full name and address of the Dutch institute and attach the form Rubicon acceptance form secondment to your application*
Name Dutch institute	
Address Dutch institute	
Employed by a foreign Host institute	☒ Yes ☐ No *If 'Yes' please attach the form Rubicon acceptance form employment to your application*

3b. Details host institute

Name of the Host institute	Champalimaud Foundation (Fundação D. Anna de Sommer Champalimaud e Dr. Carlos Montez Champalimaud)
Address	Centro de Investigação da Fundação Champalimaud, Av Brasília
Zipcode	1400-038
City	Lisbon
Country	Portugal

3c. Website of the host institute

www.fchampalimaud.org/champalimaud-research

3d. Personal details supervisor at host institute

Name supervisor	Dr. Leopoldo Petreanu
Position	Principal Investigator
Email address	leopoldo.petreanu@research.fchampalimaud.org

3e. Contacts already established with host institute

I submitted my PhD thesis for evaluation by the University of Amsterdam in March 2022, but my PhD defence did not occur until October 2022 due to scheduling conflicts of the members of the evaluation panel. During this transition period until my official graduation, in April 2022, I assumed a position as a research technician in the group of Dr. Leopoldo Petreanu, whom I had contacted prior to my thesis submission about the possibility of working in his laboratory. After my thesis defence, I officially assumed a postdoctoral position within the lab in November.
During my time as a technician, I designed the present research proposal in close consultation with Dr. Petreanu and in collaboration with Dr. Machens, to optimally match with my expertise and training opportunities, the knowledge and resources at the host institute and scientific relevance.

In 2015/2016 I stayed as a Master's student in the research group of Zach Mainen at CF for a period of 8 months.

3f. Motivation to carry out your research at host institute

After my PhD, my interest grew in understanding contextual processing beyond single areas, and more generally how areas provide the right context and information to each other. The Petreanu lab is ideally suited for this investigation given their unique expertise in optical approaches to understand cortical interactions, a topic on which Dr Petreanu has published extensively and received several international grants for. The lab has pioneered imaging activity in individual axons, cortical layers, and anatomically defined projections in combination with sophisticated approaches to probe the visual system. The extensive knowledge in the lab on the structural organization of cortical projections is really key for the aim of the current proposal - to come to a structurally informed understanding of interarea communication. This fellowship will allow learning state-of-the-art optical imaging and thus become skilled in both main neural recording methods (I used electrophysiology during my PhD).

The Petreanu lab is embedded in a world-renowned institute with top-end facilities, supported by various state-of-the-art scientific platforms. The Champalimaud Foundation is a highly dynamic, challenging and rich interdisciplinary environment for both researchers and students. Postdocs invite international scientists to give seminars and organize meetings and workshops. They receive training in writing grants and scientific communication, mentoring from Faculty, and attend an annual Postdoc retreat. The building design, shared doctorate program, internal seminars, and retreats all facilitate interactions and in-house collaborations.

The proposed population analysis of multi-area activity will be in close collaboration with the Theoretical Neuroscience lab, expert in the development and application of computational tools to understand population dynamics and circuit interactions across cortical areas. Dr Machens is an established researcher and received several international grants, including by the Simons Foundation and National Institutes of Health. The in-house collaboration involves shared lab meetings and progress meetings and thus provides an opportunity to develop theoretical understanding and multi-area analysis techniques. It further aims at an intensive dialogue between experiment and theory and this is a route I strongly believe in; we need more (empirically constrained) theory to really push our understanding of the brain.

I developed this project to specifically align with the knowledge and research lines of the Petreanu and Machens labs and with my skills. It will allow me to learn important technical and analytical skills and the institute is a remarkable place to grow as an independent researcher through the continuous interactions with experts in experimental and theoretical neuroscience.

3g. Five key items of research output

List the five key items of research output of the research group related to your research proposal at the host institute over the last five years. Do not mention H-indexes, impact factors, or any type of metric that refers to journal or publisher impact.
1. Young H, Belbut B., Baeta M., and Petreanu L* (2021). Laminar-specific cortico-cortical loops in mouse visual cortex. **ELife** 10, e59551.
2. Marques T, Nguyen J, Fioreze G, and Petreanu L* (2018). The functional organization of cortical feedback inputs to primary visual cortex. **Nat Neurosci** 21, 757–764.
3. Marques T, Summers MT, Fioreze G, Fridman M, Dias RF, Feller MB, Petreanu L* (2018). A Role for Mouse Primary Visual Cortex in Motion Perception. **Curr Biol** 28, 1703-1713.e6.
4. Broussard GJ, Liang Y, Fridman M, Unger EK, Meng G, Xiao X, Ji N, Petreanu L*, and Tian L (2018). In vivo measurement of afferent activity with axon-specific calcium imaging. **Nat. Neurosci**. 21, 1272–1280. 10.1038/s41593-018-0211-4.
5. Semedo, J.D., Zandvakili, A., Machens, C.K.*, Yu, B.M., and Kohn, A. (2019). Cortical Areas Interact through a Communication Subspace. **Neuron** 102, 249–259. 10.1016/j.neuron.2019.01.026. (note shared last three authors and that this publication reflects the proposed collaboration with Christian Machens, not the host lab)

3h. Intended starting date

Date must fall between 1 June 2023 and 1 September 2023.

01/06/2023

4. Scientific and/or societal impact of the results of the proposed project (Knowledge utilisation)

Specify which kind of impact the proposal focusses on:

Primary focus on scientific impact

Please elaborate on the scientific and/or societal impact of the proposed project:

Interactions between brain areas underlies most, if not all, brain functions and understanding the neural mechanisms underlying cortical interactions is key to understanding flexible and adaptive behaviour in humans. Knowledge of its structural basis (Aim 1) and its flexible control (Aim 2) is central to the ultimate understanding of cognition and relevant for fundamental neuroscience.

Understanding the fundamental principles of inter-area communication can have wide-ranging impact for many important research questions. The empirical data of this project are relevant for researchers studying the network dynamics of the brain at different scales, from small networks to brain-wide dynamics, but also for topics of interest to the general public, such as philosophical theories on consciousness and perception. The novelty of studying inter-areal neural communication at a population level with single-neuron structural connectivity is expected to have impact on or lead to different theoretical models of inter-area communication. These models provide the context in which to understand aspects of cognition, but also cognitive disorders. The results are therefore potentially relevant to applied research related to neurological disorders that involve disturbances in inter-areal connectivity. These include for example a misbalance between local and long-range communication in autism spectrum disorders (Khan *et al.* 2013; *PNAS*) and misbalance between bottom-up and top-down communication in schizophrenia (Fletcher and Frith 2009; *Nat. Rev. Neurosci.*). Structurally constrained models of interarea communication could yield insights that might start new avenues of study in medical research on disorders that involve aberrant inter-areal connectivity.

Besides this general impact, the proposal has several foreseeable ways in which the impact might be especially significant for the research community. First, there are various analytical techniques to estimate multivariate functional

connectivity between areas (such as Reduced Rank Regression and Canonical Correlation Analysis; Semedo *et al.* 2019, 2022). Aim 1 has the potential to show which methods align closely with structural connectivity, i.e. neurons with strong inferred connectivity are also the projection neurons. It could therefore benchmark structural inference amongst interarea functional connectivity methods, which is relevant to all researchers measuring functional connectivity without access to structural connectivity. Second, but along similar lines, the relationship between functional and structural connectivity is an established field within fMRI research. We will actively scout for possible collaboration and relevant interaction with fMRI researchers, for example the Shemesh lab at CF which focuses on ultra-high field MRI in rodents.

5. Curriculum vitae

5a. Academic Profile ⓘ

My formal training is in the neurobiological and cognitive sciences with both Bachelor and Master in neuroscience (both *cum laude*). My strongest fascination has been how neural circuit functioning relates to actual behaviour and underlies solutions to problems that organisms face in the world. Although not formally trained in computational and theoretical neuroscience, my research experience and summer schools have been at the intersection of computational and experimental approaches, including for example a computational neuroscience summer school in Okinawa, Japan (2018). My ideal is to continue research at this intersection with close connection to experiments, but strongly guided and constrained by theoretical models.

My PhD was aimed at understanding context-dependent vision in mice. I studied how the functioning of primary visual cortex (V1) is dependent on external inputs and contextual signals such as our internal state, movements, other senses and any task we are performing. This work led to four first-author publications [1-4 in Key Output section]. First, to understand how higher visual cortical areas shape processing in V1, I set up multi-area electrophysiology and optogenetics in headfixed mice and found that feedback profoundly affects sensory-evoked responses, particularly through amplification of otherwise weak sensory-evoked responses [1]. Subsequently, I aimed to understand how contextual inputs determined early visual processing under more behaviorally relevant conditions. I developed a novel audiovisual change detection task, recorded and manipulated activity across several cortical areas to study task-dependent activity and involvement of visual cortex [2], to disentangle the nature of sound-evoked activity in visual cortex [3] and study the task-dependent activity and contribution of posterior parietal cortex [4]. The data and code are publicly available for most of these first-author publications.

Whereas these studies have their particular relevance by their individual research question (see Key Output section), we obtained an integrated view of contextual interactions with early visual processing. Feedback from higher visual areas targets visually-responsive neurons, while auditory and motor signals are present in distinct subsets of neurons giving rise to a diverse palette of contextual channels to the early visual system. However, to really understand the impact of these contextual signals to visual cortex, but also from visual cortex back to other areas, we have to consider areas simultaneously. This led me to be interested in how these context-dependent interactions are implemented in the anatomy between areas and more generally how areas provide the right context and information to each other, which forms the basis of the current proposal.

My PhD work included a collaboration with two other institutes in Amsterdam (the Vrije Universiteit, and the Netherlands Institute for Neuroscience)[1] and has formed the basis for newly funded research in collaboration with the University of Amsterdam. Further initialized collaborations include Roman Rossi-Pool (UNAM University, Mexico, following a one-month research visit in March 2020), Simon van Gaal (University of Amsterdam), and Walter Senn (UNIBE, Bern, Switzerland).

Before my PhD, I worked in the lab of Zach Mainen (Champalimaud Foundation, Lisbon) on modelling serotonergic modulation of behavior in probabilistic inference paradigms and in the lab of Cyriel Pennartz on spatial coding in auditory and visual cortex. My contributions to these projects were acknowledged with multiple publications [6-8], hopefully indicative of my ability to contribute across research environments.

I further wrote articles with a clinical psychologist for a Dutch psychological trauma magazine (EMDR magazine, non peer-reviewed, 3x a year, 2017-2019) to translate neuropsychological research to clinical practice, e.g. what do brain scan results mean for treating PTSD [9]. I sincerely enjoy and value the dissemination and teaching of science. Together with a colleague, I've written a popular neuroscientific book on the brain [10]. I was a teaching assistant prior to my master's degree (4 months fulltime) and continued to teach as well as supervise student during my PhD. The act of translating and explaining science to people with different backgrounds seems to ever sharpen your own understanding as well as contextualize the relevance. I have attended more than ten national and international neuroscience conferences (FENS, SfN, CoSyNe), and have started to contribute to the larger scientific community as a reviewer for scientific journals (for example Frontiers in Systems Neuroscience).

5b. Key output ⓘ

1. **Oude Lohuis MN**, Pie JL, Marchesi P, Montijn JS, de Kock CPJ, Pennartz CMA, and Olcese U (2022). Multisensory task demands temporally extend the causal requirement for visual cortex in perception. *Nat. Commun. 13.* **"O"**

   In two multisensory detection tasks we found that visual stimuli evoked similar early sensory-evoked activity in V1, but that later activity strongly depended on whether mice were trained to report the visual stimuli. The most novel contribution of this paper is however that the neural responses, but also the causal contribution, further depended on the relevance of other sensory modalities. Adding a second modality to the task (auditory or tactile) extended the temporal window during which V1 was causally involved in visual perception.

2. **Oude Lohuis MN,** Marchesi P, Olcese U, and Pennartz CMA (2022). Triple dissociation of visual, auditory and motor processing in primary visual cortex. *BioRxiv (also under revision at Nature Neuroscience)*.

   In this paper we report that also sounds can lead to strong activity increases in primary visual cortex, but that this response was composed of two distinct components. We found a transient auditory-related component originating from auditory cortex, but also a motor-related component reflecting a rapid effect of sound-evoked movements on V1 activity. This addresses a timely issue in current multisensory research on the relevance of controlling and dissociating behavioral state-related signals from sensory inputs.

3. **Oude Lohuis MN**, Marchesi P, Pennartz CMA, and Olcese U (2022). Functional (ir)relevance of posterior parietal cortex during audiovisual change detection. *J. Neurosci. 42, 5229–5245.*

   In this publication we studied the role of Posterior Parietal Cortex (PPC) in the developed audiovisual change detection task. Despite extensive single-neuron and population-level encoding of task-relevant visual and auditory stimuli, as well as upcoming behavioral responses, optogenetic inactivation of PPC did not affect task performance.

4. **Oude Lohuis, MN,** Canton AC, Pennartz CMA, and Olcese U (2021). Higher Order Visual Areas Enhance Stimulus Responsiveness in Mouse Primary Visual Cortex. *Cereb. Cortex. bhab414.* **"O"**

   Here we describe how activity in the primary visual cortex (V1) is affected by recurrent activity originating within higher visual areas. This recurrent activity specifically enhanced stimulus responsiveness of V1 neurons by more strongly amplifying weaker compared to stronger sensory-evoked responses and in a similar manner during anesthetized as awake conditions.

5. Olcese, U, **Oude Lohuis MN**, and Pennartz CMA (2018). Sensory Processing Across Conscious and Nonconscious Brain States: From Single Neurons to Distributed Networks for Inferential Representation. *Front. Syst. Neurosci. 12 (49).*

   In this review, we describe how global and local brain states impact sensory processing and integrate both rodent and human work with theoretical ideas of why sensory processing during sleep or anaesthesia renders it unconscious. It has been cited by diverse fields of research.

6. Vertechi P, Lottem E, Sarra D, Godinho B, Treves I, Quendera T, **Oude Lohuis MN**, and Mainen ZF (2020). Inference-Based Decisions in a Hidden State Foraging Task: Differential Contributions of Prefrontal Cortical Areas. *Neuron 106:166-176e6.* **"O"**

This work stemmed from my master's internship in the lab of Zach Mainen. I contributed to the initial design of the normative formulation of the paradigm, programming of the behavioural task and mouse behavioural experiments.

7. Lottem E, Banerjee D, Vertechi P, Sarra D, **Oude Lohuis MN**, and Mainen ZF (2018). Activation of serotonin neurons promotes active persistence in a probabilistic foraging task. *Nat. Commun. 9, 1000.* **"O"**

This publication resulted from my period in the lab of Zach Mainen, similar to [6]. I performed optogenetic manipulations of serotonergic activity in the dorsal raphe nuclei of the brainstem. This article is well-cited for helping to disambiguate different theories of serotonin function (behavioral inhibition versus inference and uncertainty control).

8. Mertens, PE, Marchesi P, **Oude Lohuis MN**, Krijger Q, Pennartz CMA, and Lansink CS (2021). Coherent mapping of position and head direction across auditory and visual cortex. *BioRxiv (also under revision at Cerebral Cortex)*

I provided contribution to this project during my first master's internship at the University of Amsterdam. Amongst others I helped setting up and executing behavioural training.

9. **Oude Lohuis MN** & Hornsveld HK (2020) EMDR bij muizen...?! *EMDR magazine 20, 9-11*. (www.emdr-magazine.nl/emdr-magazine-nr-20/)

This article is an illustration of a series of articles written with a clinical psychologist aimed at translating research on trauma therapies (in particular Eye Movement Desensitization and Reprocessing, or EMDR) for Dutch clinicians using EMDR therapy. In this article, we discuss the implications of research with mice to understand the working mechanisms of EMDR therapy.

10. Geluk H & **Oude Lohuis MN** (2015) Hersenen voor in bed, op het toilet of in bad. Popular scientific book on the most fascinating neuroscience topics (in Dutch). *BBNC Uitgevers, ISBN 9789045318431.*

Public outreach is important to justify science and I personally love to highlight the amazing properties of our most important organ and make neuroscientific stories accessible and entertaining for the wider public. Together with my friend and colleague Han Geluk I wrote a popular scientific book (in Dutch) on the curiosities of neuroscience.

5c. Identify most important output indicators within your field of research

The primary output indicator of my field are scientific articles published in peer-reviewed journals. In the field of cognitive and systems neuroscience these publications often involve substantial effort to obtain high-quality neural recordings during animal behaviour and subsequent analyses to understand this high-dimensional data. Indicators involve the number of peer-reviewed scientific publications, their associated journal and impact factor and the number of citations they receive or times downloaded.

Further indicators of research quality are whether the research is being followed up on, has formed the basis for new research (grants) and whether it has led to successful collaborations, all indicative of scientific interest by others. A high number of invited seminars or attendances to scientific conferences further show that the work is valued externally as many conferences require high-quality abstracts or even have peer-reviewed selection with low acceptance rates (for example the CoSyNe meeting).

Additional publications such as books, book chapters and reviews show that the applicant has larger interest and the ability to oversee and integrate the latest research.

5d. Possible impact of the Rubicon project on your personal career

My career plans are to attain a permanent faculty position, preferably in the Netherlands. A granted Rubicon fellowship would be extremely valuable to work towards this goal. This postdoctoral research project involves training in cutting

edge technologies and computational methods that complement my PhD training. Moreover, the competitive environment of high scientific level at CF allows me to grow as an independent researcher at an international institute and having two world-leading scientists as supervisors provides access to a large professional network. As I aspire to return to the Netherlands for a permanent research position after my post-doctoral phase, having received a Rubicon grant will probably increase the likelihood of being able to obtain further funding.

The current proposal has two somewhat more specific benefits. The collaboration and training in theoretical neuroscience will broaden my methodology and possible research lines. Second, I expect the resultant dataset of the current proposal to be rich (given the large-scale multi-area imaging) and potentially informative on questions beyond those in this proposal. I envisage future collaboration given that I value translational neuroscience and the relationship between structure and function in network dynamics is a ubiquitous issue in neuroscience research ranging from lower organisms to humans.

6. Administrative details

6a. Personal details

Academic title (e.g. Dr or MA/MSc):	Dr.
Initial(s):	M.N.
Prefix:	
Surname	Oude Lohuis
Preference for correspondence in English:	☐ Yes ☒ No
Address for correspondence	R. Sao Joao da Mata 35-1
Zip code:	1200-850
City:	Lisbon
Country:	Portugal
Email address:	matthijsoudelohuis@gmail.com
Telephone number (mobile):	+31634312649

6b. Master's ('doctoraal')

University/College of Higher Education:	University of Amsterdam
Faculty:	Swammerdam Institute for Life Sciences
Date of certificate:	22-12-2016
Main subject:	Basic and Applied Neuroscience

6c. Doctorate ('promotie')

University/College of Higher Education	University of Amsterdam
Date and place of (intended) doctoral degree as stated on your certificate:	06-10-2022
Supervisor ('Promotor'):	Prof. Dr. Cyriel M.A. Pennartz
Title of thesis:	Contextual signals in visual cortex: how sounds, state, and task setting shape how we see

6d. Extension clause

I have permission to make use of the extension clause

☐ Yes

☒ Not applicable

6e. Current employment and work experience since graduating (see 6b) ⓘ

	Fixed term	Permanent position	Full time equivalent	Start date	End date	Employer	Place	Scientific discipline
PhD student	☒ Yes ☐ No	☐ Yes ☒ No	1 FTE	Date 01/01/2017	Date 31/12/2021	University of Amsterdam	Amsterdam, NLD	Neuroscience
Other	☒ Yes ☐ No	☐ Yes ☒ No	1 FTE	Date 01/04/2022	Date 31/10/2022	Champalimaud Research	Lisbon, Portugal	Neuroscience
Postdoc	☐ Yes ☒ No	☐ Yes ☒ No	1 FTE	Date 01/11/2022	Date 01/06/2023	Champalimaud Research	Lisbon, Portugal	Neuroscience

6f. Months spent on research ⓘ

Employment history

Dec 2016: M.Sc. Graduation

I. May 2013 – January 2020: Dutch Brain Bank. 0.1 FTE position. 100% to be spent on administrative tasks. Night/Weekend shifts on autopsy team, organizing and facilitating sectioning after passing of a brain donor.

II. January 2017 – December 2021: University of Amsterdam. 1 FTE position. 90% to be spent on research, 10% on education.

III. January 2022 – March 2022: unemployed.

IV. April 2022 – October 2022: Champalimaud Research. 1 FTE position. 100% to be spent on research.

V. November 2022: Champalimaud Research. 1 FTE position. 100% to be spent on research.

Calculation months of research

*I. 36 months (since Jan 2017) * 0.1 FTE position * 1.0 spent on administrative tasks = 4 months*

*II. 60 months * 1 FTE position * 0.9 spent on research = 54 months*
*60 months * 1 FTE position * 0.1 spent on education = 6 months*

*III. 7 months * 1 FTE position * 1.0 spent on research = 7 months*

*IV. 1 months * 1 FTE position * 1.0 spent on research = 1 months*

Experience	Number of months
Research activities	(54 + 7 + 1) = 62
Teaching	6
Care or sick leave	0
Administrative tasks	4
Others (please specify):	3 (unemployed, finishing PhD work)

7. Appendices and word count

Please submit the application and the appendices to NWO/ZonMw in electronic form (pdf format is required!) using the online application system ISAAC (NWO)/ProjectNet (ZonMw), which can be accessed via the NWO/ZonMw website (www.nwo.nl and www.zonmw.nl).

Do not forget to attach the:

- two recommendation forms (mandatory)
- the applicable acceptance form (signed) (mandatory)
- the manuscript approval form (when applicable).

Recommendation forms

Give the title(s), initial(s), surname and university of the two references expressing support for the application on the two recommendation forms accompanying the application.

Reference 1		Reference 2	
Title(s):	Prof. Dr.	Title(s):	Dr.
Initial(s):	C.M.A.	Initial(s):	U.
Surname:	Pennartz	Surname:	Olcese
University:	University of Amsterdam	University:	University of Amsterdam

The recommendation forms should preferably be submitted by the applicant through the ISAAC/ProjectNet system but if the authors of the recommendations have objections to that procedure, they may send their forms by email to rubicon@nwo.nl **before the applicable deadline**. Fill in the box, choose yes or no.

The forms are attached to this application:	☒ Yes ☐ No
The forms will be sent by email to rubicon@nwo.nl:	☐ Yes ☒ No

Please note that when submitting your application through ProjectNet, the system requests a confirmation that you have submitted your application. **You don't need to submit this confirmation for the Rubicon programme**.

Word count

	Exact number of words for each of the paragraphs:	Maximum number of words allowed:
1b	**199**	200
1f NL	**44**	50
1f EN	**50**	50
2a	**1987**	2000
3e	**161**	200
3f	**398**	400
4a-d	**388**	400
5a	**682**	700
5b	**519**	700
5c	**170**	200
5d	**200**	200

Submit application

Before submitting your application please check and make sure that:
- your gender and first name(s) are not mentioned in your application
- active hyperlinks are not in your application, except for the Key Output (section 5b).

- the Explanatory Notes are not visible in the PDF version of your application. Please remove the General Explanatory Notes, the general notes on section 1, the general notes on section 2, the general notes on section 4, the general notes on section 5, the general notes on section 6

Submit the application through: http://www.nwo.nl/calls/rubicon-2022-3 via the appropriate window before the deadline expires. The deadline is November 29, 2022 at 14:00:00 hours CEST (2 PM sharp).